

ML and DS Internship Practical

Month 1 – Python Fundamentals

Student Management System

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Submitted To: -

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Technology Used: Python

Development Environment: Visual Studio Code

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1 Objective: -

The objective of this project is to develop a console-based Student Management System using Python. The application enables users to manage student records efficiently by performing operations such as adding, viewing, searching, updating, and deleting student information.

2 Introduction: -

The Student Management System is a menu-driven Python application designed to manage student records. It allows users to store and maintain student details in an organized manner. The project demonstrates the practical implementation of Python fundamentals and provides hands-on experience with data management operations.

3 Features: -

1. Add Student Record
2. View All Student Records
3. Search Student by Student ID
4. Update Student Information
5. Delete Student Record
6. Input Validation
7. Menu-Driven Interface
8. User-Friendly Console Output

Input Validation Features

The system ensures that all required fields are entered before saving or updating student records.

- If Student ID is empty, the message "Please enter Student ID" is displayed.

- If Student Name is empty, the message "Please enter Student Name" is displayed.
- If Course is empty, the message "Please enter Course" is displayed.
- If Marks are empty, the message "Please enter Marks" is displayed.
- If Marks are entered $100 >$, the message "Marks must be between 0 and 100. Please Enter Valid Marks."

4 Concepts Used: -

Variables

Used to store student details such as ID, name, course, and marks.

Data Types

- Integer (Student ID)
- String (Student Name, Course)
- Float (Marks)

Input and Output Operations

Used to receive user input and display results.

Conditional Statements

Used for decision-making and validating user inputs.

Loops

- while loop for displaying the menu repeatedly.
- for loop for traversing student records.

Functions

Functions used in the project:

- add_student()
- view_students()

- search_student()
- update_student()
- delete_student()

Lists

A list is used to store multiple student records.

Dictionaries

Each student record is stored as a dictionary containing student information.

Input Validation

Validation ensures that all mandatory fields are entered correctly before processing.

CRUD Operations

- Create → Add Student
- Read → View/Search Student
- Update → Update Student Details
- Delete → Delete Student Record

Menu-Driven Programming

The application operates through a menu that allows users to select different actions.

5 Project Workflow: -

1. The application displays the main menu.
2. The user selects an operation.
3. The system performs the requested action.
4. Student records are stored and managed using lists and dictionaries.

5. The menu is displayed again until the user exits the program.

6 Learning Outcomes: -

Through this project, the following concepts were learned:

- Python Syntax
- Variables and Data Types
- Conditional Statements
- Loops
- Functions
- Lists
- Dictionaries
- Input Validation
- CRUD Operations
- Problem Solving Skills

7 Advantages: -

- Easy to use
- Beginner-friendly
- Improves programming skills
- Prevents incomplete student records
- Enhances logical thinking

8 Limitations: -

- Data is stored temporarily in memory.
- Records are lost when the application is closed.
- No database integration.

9 Testing Report: -

Test Case ID	Test Description	Test Input	Expected Result	Status
TC01	Add Student with valid data	ID=101, Name=Saniya, Course=BCA, Marks=85	Student added successfully	Pass
TC02	View student records	Select View Option	All student records displayed	Pass
TC03	Search existing student	ID=101	Student found	Pass
TC04	Search non-existing student	ID=999	Student not found message displayed	Pass
TC05	Update existing student	ID=101	Student details updated successfully	Pass
TC06	Delete existing student	ID=101	Student deleted successfully	Pass
TC07	Search deleted student	ID=101	Student not found message displayed	Pass
TC08	Enter empty Student ID	Blank Input	"Please enter Student ID" displayed	Pass
TC09	Enter empty Student Name	Blank Input	"Please enter Student Name" displayed	Pass

TC10	Enter empty Course	Blank Input	"Please enter Course" displayed	Pass
TC11	Enter empty Marks	Blank Input	"Please enter Marks" displayed	Pass
TC12	Enter alphabet in Student ID	abc	Student ID must be contain numbers only	Pass
TC13	Enter special characters in Student ID	@#\$	Student ID must be contain numbers only	Pass
TC14	Enter negative marks	-10	"Marks must be between 0 and 100.Please Enter Valid Marks"	Pass
TC15	Enter marks greater than 100	120	"Marks must be between 0 and 100.Please Enter Valid Marks"	Pass
TC16	Enter invalid menu choice	9	"Invalid Choice" displayed	Pass
TC17	Add multiple students one by one	3 Student Records	All records stored successfully	Pass
TC18	Update non-existing student	ID=999	Student not found message displayed	Pass
TC19	Delete non-existing student	ID=999	Student not found message displayed	Pass

TC20	Exit application	Option 6	Program terminated successfully	Pass
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10 Screenshot: -

Menu Options

```

PS C:\Users\Saniya Shaikh> "C:/Users/Saniya Shaikh/AppData/Local/Programs/Python/Python3
===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice:

```

1 Add Student process succesfully

```

===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice: 2

Student Records:

ID      :1
Name    :Saniya
Course  :BCA
Marks   :85.0

```

2 View Student

```

PS C:\Users\Saniya Shaikh> "C:/Users/Saniya Shaikh/AppData/Local/Programs/Python/Python3
===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice: 1
Enter Student ID: 1
Enter Name: Saniya
Enter Course: BCA
Enter Marks: 85
Student Added Successfully!

```

3 Search Student

```
===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice: 3
Enter Student ID to Search: 1
Student Found:
ID      :1
Name    :Saniya
Course  :BCA
Marks   :85.0
```

4 Update Student

```
===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice: 4
Enter Student ID to Update: 1
Enter New Name: Sanu
Enter New Course: BBA
Enter New Marks: 79
Student Updated Successfully
```

5 Delete Student

```
===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice: 5
Enter Student ID to Delete: 1
Student Deleted Successfully!
```

6 Exit

```
===== Student Management System =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter Your Choice: 6
Thank You!
```

11 Conclusion: -

The Student Management System successfully demonstrates the implementation of Python fundamentals including variables, data types, loops, functions, lists, dictionaries, and input validation. The project provides practical experience in developing a menu-driven application and performing CRUD operations. It also improves problem-solving skills and serves as a strong foundation for more advanced software development projects.

12 GitHub Repository Link: -

<https://github.com/saniya2698/Python-Student-Management-System>